



Effect of different levels of clove (*Syzygium aromaticum* L.) essential oil on growth performance and oxidative/nitrosative stress biomarkers in broilers under heat stress

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Abstract

A 49-day fully randomized trial was conducted to investigate the dietary effects of clove (*Syzygium aromaticum* L.) essential oil (CEO) on growth performance and oxidative/nitrosative stress biomarkers in broilers under heat stress. A total of 288 male broilers (Ross 308) were randomly divided into 6 dietary groups (4 replicates and 12 birds/replicate) and supplemented as follows: (I) Normal control (NC) received only basal diet under normal condition. The rest of the animals were challenged with heat and assigned to the following groups: (II) Heat stress control (HSC) received only basal diet; (III) Standard treatment (ST) received basal diet + vit E (100 ppm); (IV–VI) Herbal treatments (HT) received basal diet + 250, 350, and 450 ppm CEO. Heat stress could significantly decrease the animals' performance and induce severe oxidative/nitrosative stress. The HT at the middle dose could significantly improve body weight, body weight gain, and feed intake compared to HSC; however, none of the treatments had a significant effect on feed conversion ratio after inducing heat stress. Moreover, both ST and HT with a trend towards concentration-dependent fashion significantly contributed to normalization of oxidative/nitrosative biomarkers. It appears that CEO is a potential replacement for synthetic antioxidants in broiler diets.

Keywords *S. aromaticum* · Oxidative stress · Broiler · Heat stress · Clove

Introduction

Heat stress is one of the inevitable challenges of poultry industry particularly in tropical and subtropical countries due to the global increase in environmental temperature. The thermal injury is commenced when the ambient environmental temperature goes beyond the thermo-neutral zone (16–25 °C). It is a well-known concept that heat stress can negatively influence growth and production of animals, as well as induce metabolic alterations and oxidative stress (Abd El-Hack et al. 2020). The most favorable and practical way to alleviate the heat stress-induced alterations is perhaps dietary modifications by the use of commercial feed additives. In fact, these additives are routinely used in poultry industry to achieve a variety of purposes, most likely enhance growth performance, immunity, and antioxidant status (Al-Sultan et al. 2019). Despite their

beneficial effects, these additives may come with major side effects to both animals and humans, in terms of residues in food. With this aspect, researchers are continuously striving to replace the current synthetic additives with those of natural, safe, efficient, cheap, and readily available ones. Among various options, herbal materials have aroused considerable research interest and various herbal derivatives including dried powder, extract, and essential oil (EO) are currently being explored in poultry industry. In addition to the aforementioned benefits, herbal supplements are versatile, meaning that they can impose several beneficial effects on chickens simultaneously, upon administration. For instance, Hussein et al. (2018) demonstrated that dietary supplementation with cold-pressed clove (*Syzygium aromaticum* L.) oil could enhance growth performance, improve health status, and reduce intestinal pathogens in Japanese quails (Hussein et al. 2018).

Birds' performance can be undermined by oxidative stress induced under stressful conditions such as heat exposure or coccidiosis, due to overproduction of reactive oxygen species (ROS) such as superoxide, hydroxyl, and hydrogen peroxide radicals (Dalloul et al. 2006). In normal conditions, both

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enzymatic and non-enzymatic antioxidant systems work together to eliminate excessive oxidative radicals. In fact, the process is tightly regulated keeping oxidants and antioxidants in balance. However, when the oxidant molecules are overproduced, they violate the capacity of antioxidants leading to a chain of adverse events including lipid peroxidation, protein carbonylation, and DNA destruction (Nazarizadeh and Asri-Rezaie 2016). Oxidative status is mainly assessed by measuring total antioxidant capacity (TAC), malondialdehyde (MDA), carbonylated protein (PCO), and activities of three key antioxidant enzymes including superoxide dismutase (SOD), glutathione peroxidase (GSH-Px), and catalase (CAT). Moreover, the status of nitrosative stress can be evaluated by measuring nitric oxide (NO) in biological samples (Wang et al. 2008). Supplementation with synthetic antioxidants (e.g., α -tocopheryl acetate or butylated hydroxytoluene) has been proposed as a practical way to alleviate oxidative damages in the poultry industry. However, recently the global trend has been changed and researchers have concentrated on herbal materials.

S. aromaticum belonging to the family Myrtaceae is thought to be one of the most versatile herbs. Clove is a valuable source of vitamins (vit) including A (retinol); beta-carotene; and vits K, B, and C (Al-Mufarrej et al. 2019). Due to the presence of several bioactive components, including flavonoids, hydroxybenzoic acids, hydroxycinnamic acids, and hydroxyphenyl propane, clove has antioxidant, antimicrobial, antinociceptive, and cytotoxic effects (Cortés-Rojas et al. 2014). Particularly, clove essential oil (CEO) and its main constituent, eugenol, have been reported to possess useful analgesic, anesthetic, and antiseptic effects (Batiha et al. 2020). Essential oils (EOs) refer to a mixture of aromatic oily liquids derived from plant materials such as flowers, buds, seeds, leaves, twigs, bark, herbs, wood, fruits, and roots. Owing to their versatile nature, EOs have been practiced for their role as antibacterial, antiviral, antifungal, antioxidant, digestive stimulant, immunomodulator, a hypolipidemic agent, and also heat stress alleviator in animal feeding (Gopi et al. 2014).

Different forms of clove including dried powder, extract, and EO have previously been reported to impose beneficial effects on growth performance and production of both broilers and laying hens (Al-Mufarrej et al. 2019, Alizadeh et al. 2015, Dalkiliç and Güler 2009, Gandomani et al. 2014, Hussein, Abd El-Hack, Mahgoub, Saadeldin and Swelum 2018, Mehr et al. 2014, Mohammadi et al. 2014). However, all of the previous studies were conducted under normal condition. Additionally, despite the importance of oxidative/nitrosative stress biomarkers, none of these studies evaluated them or at least the performed studies are very limited and insufficient (Mahrous et al. 2017; Petrovic et al. 2012). Therefore, the current research aimed at the abovementioned shortcomings and was undertaken to investigate the effects of different

levels of CEO on growth performance and oxidative/nitrosative stress biomarkers in broilers under heat stress. To the authors' knowledge, this is the first study done in the field which tests the possible beneficial effects of CEO on oxidant/antioxidant status compared to a commercial antioxidant supplement (vit E).

Materials and methods

Experimental design and dietary treatments

The experiment was conducted at the Experimental Livestock Unit of Azad University, Kermanshah Branch, Iran. A total of 288 uniform 1-day-old male broiler chicks (Ross 308) with an average weight of 40.86 g were purchased from a local commercial hatchery and randomly allocated to one of the 6 major groups. Each main group had 4 replicates with 12 chicks in each replicate. The birds were kept in pens with an area of $150 \times 120 \text{ cm}^2$ and one hanging tube feeder and one suspended drinker. The birds were reared on concrete flooring covered with wood shavings as litter material. The chicks assigned to the normal control (NC) group were separated from the rest of the birds and housed in another room with a normal temperature. All of the animals had free access to food and water during the experimental course (49 days). Nutritional requirements of the chickens were met using three different corn-soybean meal diets for the starter (days 1–14), grower (days 15–29), and finisher (days 30–49) periods based on the NRC guidelines (Nutrition 1971). Further details can be found in Table 1. Trace mineral and vit premixes (0.25%) were added to the basal diet, but no commercial antioxidant was included. The dietary treatments were (I) normal control (NC), in which the chicks of this group were kept under recommended condition and received the basal diet; (II) heat stress control (HSC), in which the animals of this group were challenged with heat and received the basal; (III) standard treatment (ST), in which the animals of this group were challenged with heat and received the basal supplemented with 100 ppm of vit E (DL- α -tocopherol acetate) as a standard commercial antioxidant; (IV–VI) herbal treatment (HT), in which the animals of these groups were challenged with heat and received the basal diet supplemented with 250, 350, and 450 ppm of CEO. The herbal material was thoroughly mixed with the basal diet. The selected dose of CEO was based on the previous study (Mohammadi et al. 2014). Vit E (Microvit™ E Promix 50) was procured from a local commercial company (Saral Poultry Feed Supplementation, Kermanshah, Iran). The selected dose of vit E was based on the previous study (Habibi et al. 2014) and our pilot studies (unpublished data). The lighting schedule was 23-h light/1-h dark and the temperature was kept at 32 °C for the first 2 days; then, it was gradually reduced by 0.5 °C per day until 13 days

Table 1 Ingredient and nutrient composition of the basal diets

	Starter	Grower	Finisher
Composition (g/kg DM)			
Corn	541.6	612.9	559.3
Wheat	0	0	70
Soybean meal (45%)	393.2	308.9	278.9
Soybean oil	25.1	35.2	50.7
L-Lysine hydrochloride	1.6	1.7	2.0
DL-Methionine	3.4	2.8	2.9
D-Calcium phosphate	15.1	19.3	17.8
Oyster shell	11.2	10.3	9.8
Common salt	3.7	3.9	3.6
Mineral mixture*	2.5	2.5	2.5
Vitamin mixture**	2.5	2.5	2.5
Calculated nutrient content (g/kg DM)			
Metabolic energy (MJ/kg)	12.30	12.89	13.28
Crude protein	222.6	190.1	180.1
Lysine	13.2	11.2	10.5
Methionine	6.49	5.50	5.42
Methionine + cysteine	9.8	8.4	8.2
Calcium	9.1	9.6	9.1
Phosphorus, available	4.2	4.8	4.5
Sodium	1.6	1.7	1.6
Anion-cation balance (meq/kg)	235.17	196.9	184.07

*Trace mineral mix provided (mg/kg of diet): $\text{MnSO}_4 = 248$ mg; $\text{FeSO}_4 \cdot 7\text{H}_2\text{O} = 125$ mg; $\text{ZnO} = 211$ mg; $\text{CuSO}_4 = 25$ mg; $\text{CaI}_2 = 25$ mg; $\text{Se} = 0.5$ mg; choline = 625 mg

**Supplied per kilogram of diet: retinyl acetate = 22,500 IU; cholecalciferol = 5000 IU; α -tocopherol = 45 IU; thiamine = 4.3 mg; riboflavin = 16.5 mg; cobalamin = 0.04 mg; pantothenic acid = 24.5 mg; folic acid = 2.5 mg; niacin = 74 mg; pyridoxine = 7.3 mg; biotin = 40 mg

of age. The relative humidity in the room was maintained at the range of 45–65% (Habibi et al. 2014). Animal care was performed according to the recommendations in the Guide for the Care and Use Committee of Islamic Azad University, Kermanshah Branch, Iran.

Heat stress

From day 14 until 42, the ambient temperature was adjusted so that daily fluctuations were between 21.9 and 36 °C, simulating diurnal temperature during heat stress. The birds were exposed to 12 h of 21.9 °C, 3 h of 21.9 to 36 °C, 5 h of 36 °C, and 4 h of 36 to 21.9 °C. The temperature cycle was scheduled based on the previous study with slight modifications (Akbari and Torki 2014). From day 42 until the end of the experimental period (day 49), chickens were maintained at the recommended comfort temperature (19–22 °C).

Plant material

CEO was purchased from Barij Essence (Kashan-Iran). The company provided the gas chromatography/mass spectroscopy (GC/MS) analysis of the essential oil as presented in Table 2.

Assessment of growth performance

The live body weight (BW) was measured at 10-, 22-, 42-, and 49-day intervals of the experiment and body weight gain (BWG) was calculated for 1–10-, 11–22-, 23–42-, and 43–49-day periods. Feed consumption (FC) was recorded in the aforementioned periods and feed conversion ratio (FCR) was calculated after correcting for mortalities (FI/BWG) (Habibi et al. 2014).

Sampling

At the endpoint, 6 randomly chosen broilers were slaughtered by decapitation in each replicate (24 out of 48) and the blood specimens were collected into heparinized test tubes, then centrifuged for separation of plasma samples at 1180 g for 15 min. The blood and plasma samples for the analysis of antioxidant indices were frozen and stored at – 65 °C (Petrovic, Marcincak, Popelka, Simkova, Martonova, Buleca, Marcincakova, Tuckova, Molnar and Kovac 2012).

Assessment of oxidative/nitrosative stress parameters

All of the indicators of oxidative/nitrosative stress were analyzed using an automated biochemistry analyzer (BT1500, Biotechnica, Italy). The biomarkers were measured using standard assay kits and according to the previous study (Nazarizadeh and Asri-Rezaie 2016).

Very briefly, the activities of the antioxidant enzymes were measured spectrophotometrically using standard assay kits of CAT (Cayman, USA, Item No. 707002), SOD (RanSod,

Table 2 Chemical composition of clove essential oil determined by GC-MS

No.	compound	Concentration (%)	Retention time (min)
1	Eugenol	73.19	12.73
2	Caryophyllene	5.73	13.91
3	Eugenol acetate	9.57	13.32
4	Cycloundecatriene	8.04	15.09
5	Toluene	1.44	3.59
Total 98%			

RanDox Co., UK), and GSH-Px (Ransel, RanDox Co., UK). The activities of the aforementioned enzymes were expressed as U/g Hb. Since the units are according to hemoglobin (Hb) contents of erythrocytes, the Hb levels were measured using Drabkin's reagent.

Lipid peroxidation was determined as thiobarbituric acid reactive substance (TBARS) according to the previous method (Buege and Aust 1978) with trifling alterations. The assay was done using trichloroacetic acid, thiobarbituric acid, and L-1 hydrochloric acid. The total antioxidant capacity was determined using ABTS (2,2'-azino-di-[3-ethylbenzthiazoline sulfonate]; Randox Laboratories Ltd., UK) as the substrate. The total nitrate/nitrite content of plasma samples was measured according to the routine method of Griess reaction. Briefly, 100 µl of the sample was mixed with 100 µl of Griess reagent in a 96-well plate and the reaction was read at 570 nm after 10 min (Ding et al. 1988). PCO was measured based on the method of Levin and collaborators (Levine et al. 1990), using 2,4-dinitrophenylhydrazine, hydrochloric acid, ethanol-ethyl acetate (1:1, v/v), and guanidine hydrochloride solution (6 M). The carbonyl content was calculated based on the molar extinction coefficient of DNPH ($\varepsilon = 2.2 \times 10^4$ cm/M).

Statistical analysis

The packaged SPSS program for windows (Version 22. 2013, Chicago, IL, USA) was used for statistical analysis. Data were expressed as mean and standard deviation (means $\pm$ SD). The data normality was assessed using the Shapiro-Wilk test and graphical assessment (histogram and Q-Q plot). The differences between groups were determined by analysis of variance (ANOVA) followed by the pair-wise comparison Bonferroni test. $P < 0.05$ was considered statistically significant.

Results

The effect of dietary supplementation with vit E and different levels of CEO on growth performance of broilers during the feeding trial (49 days) is shown in Table 3. As can be seen, heat stress significantly reduced the animals' performance compared to those raised under normal condition (NC group). Only supplementation with the middle dose of CEO (350 ppm) could significantly ($P < 0.05$) increase BW during all the time points compared to the HSC group. Additionally, the lowest and highest doses of CEO (250 and 450 ppm) had no significant difference with ST. At the first interval (1–10 days), HT with the doses of 350 and 450 ppm could significantly increase BWG compared to the NC group. HT also could stand against heat stress and significantly improved BWG at the second interval (11–22) compared to the HSC group; however, the middle dose (350 ppm) of HT acted better than the other doses.

At the third interval (23–42), only the middle dose of CEO could significantly enhance BWG compared to HSC. The middle dose also could significantly elevate the overall BWG, but the other treatments had no significant effect on the final BWG compared to HSC. FI was significantly increased in the birds treated with 350 ppm of CEO at the third interval (23–42) comparable to those of the NC group. Moreover, the middle dose of CEO could also significantly increase the overall FI (day 49) compared to HSC. As indicated in Table 3, none of the treatments could significantly alter FCR compared to HSC after inducing heat stress. However, before beginning of the heat stress (the first interval), the middle dose of HT significantly decreased FCR compared to the other groups.

Table 4 represents the results obtained from assessing oxidative/nitrosative stress biomarkers in the broilers under normal condition (NC) and heat stress. As can be seen, heat stress could significantly increase the activities of all the three enzymes and the MDA, PCO, and NO levels as well as decrease TAC levels compared to the NC group. Supplementation with both the vit E and CEO could significantly contribute to the normalization of the biomarkers except for SOD. To be specific, the activity of SOD was always high and none of the treatments could significantly reduce it. On the other side, CEO at the doses of 350 and 450 ppm significantly decreased GSH-Px activity compared to HSC. The activity of CAT was also significantly declined by both ST and HT compared to HSC; however, the highest dose of CEO (450 ppm) had a more pronounced effect compared to the other treatments. Moreover, the treatments could also protect lipids and proteins against oxidative damage, as evidence by both MDA and PCO levels which were significantly decreased following supplementation with vit E and CEO compared to the HSC group. Interestingly, it was observed that the highest dose of CEO (450 ppm) could reduce MDA and PCO levels comparable to that of the NC group. NO content was not significantly influenced by vit E and 250 ppm of CEO compared to HSC. However, the middle and highest doses of CEO could significantly reduce NO content with a tendency towards dose-dependent fashion. TAC levels were restored by both ST and HT in a dose-dependent fashion and the highest dose of CEO was able to enhance TAC close to the normal range, as there was no significant difference between 450 ppm of CEO and NC.

Discussion

As mentioned above, clove was previously tested in both laying hens and broilers for possible beneficial effects; however, all of the previous studies were conducted under normal condition, without any stressor, and none of these studies

Table 3 Effects of different levels of clove essential oil (CEO) and vit E on body weight (BW), body weight gain (BWG), feed intake (FI), and feed conversion ratio (FCR) in broilers

Parameter	Period (day)	NsC	HSC	100 ppm vit E	250 ppm CEO	350 ppm CEO	450 ppm CEO
BW (g)	10	224.75 ± 3.11 ^a	226.19 ± 4.29 ^a	229.12 ± 5.19 ^a	230.23 ± 4.27 ^a	244.86 ± 3.14 ^b	239.18 ± 5.71 ^{ab}
	22	867.53 ± 7.29 ^a	658.11 ± 7.63 ^b	663.69 ± 8.21 ^b	673.71 ± 9.62 ^b	705.16 ± 7.44 ^c	680.51 ± 10.39 ^{bc}
	42	2191.14 ± 24.12 ^a	1758.92 ± 26.12 ^b	1793.18 ± 20.82 ^b	1799.92 ± 19.64 ^b	1853.92 ± 12.16 ^c	1796.92 ± 18.78 ^b
	49	2862.58 ± 12.53 ^a	2325.31 ± 18.52 ^b	2359.31 ± 19.68 ^b	2358.93 ± 16.38 ^b	2418.77 ± 16.24 ^c	2368.77 ± 20.54 ^b
	1–10	184.56 ± 2.91 ^a	186.32 ± 3.71 ^{ac}	189.12 ± 3.99 ^{ac}	190.42 ± 4.05 ^{ac}	205.17 ± 3.39 ^b	198.66 ± 4.83 ^{bc}
BWG (g)	11–22	643.78 ± 7.51 ^a	426.29 ± 4.33 ^b	432.16 ± 4.82 ^b	445.36 ± 3.09 ^c	461.98 ± 5.12 ^d	443.79 ± 4.73 ^c
	23–42	1324.29 ± 25.11 ^a	1106.24 ± 30.18 ^b	1130.39 ± 22.71 ^b	1124.52 ± 25.82 ^b	1198.86 ± 19.58 ^c	1121.33 ± 16.79 ^b
	43–49	671.93 ± 8.34 ^a	567.21 ± 6.93 ^b	566.44 ± 8.42 ^b	559.77 ± 7.12 ^b	565.97 ± 4.37 ^b	571.63 ± 9.68 ^b
	1–49	2821.24 ± 19.06 ^a	2284.29 ± 18.89 ^b	2318.87 ± 16.47 ^b	2317.12 ± 17.38 ^b	2377.83 ± 20.18 ^c	2327.66 ± 22.93 ^b
FI (g)	1–10	245.98 ± 3.51 ^a	251.19 ± 4.07 ^a	248.93 ± 4.15 ^a	251.73 ± 5.08 ^a	241.64 ± 3.97 ^a	253.08 ± 5.39 ^a
	11–22	981.33 ± 6.28 ^a	782.59 ± 7.43 ^b	788.18 ± 6.93 ^b	797.65 ± 7.69 ^b	806.75 ± 8.92 ^b	789.92 ± 5.03 ^b
	23–42	2518.21 ± 21.36 ^a	2431.08 ± 16.53 ^b	2439.58 ± 17.49 ^b	2430.52 ± 22.97 ^b	2538.79 ± 28.14 ^a	2402.67 ± 26.79 ^b
	43–49	1301.58 ± 15.21 ^a	1166.79 ± 16.95 ^b	1175.21 ± 14.54 ^b	1169.42 ± 16.37 ^b	1165.37 ± 14.79 ^b	1161.45 ± 15.37 ^b
	1–49	5239.42 ± 23.14 ^a	4873.44 ± 27.89 ^b	4893.17 ± 29.32 ^b	4908.83 ± 24.92 ^b	4996.11 ± 22.71 ^c	4914.79 ± 28.44 ^b
FCR (g/g)	1–10	1.33 ± 0.04 ^a	1.34 ± 0.03 ^a	1.30 ± 0.02 ^a	1.30 ± 0.02 ^a	1.18 ± 0.03 ^b	1.28 ± 0.03 ^a
	11–22	1.53 ± 0.03 ^a	1.79 ± 0.05 ^b	1.77 ± 0.05 ^b	1.77 ± 0.04 ^b	1.75 ± 0.02 ^b	1.76 ± 0.03 ^b
	23–42	1.90 ± 0.04 ^a	2.18 ± 0.04 ^b	2.15 ± 0.05 ^b	2.16 ± 0.05 ^b	2.12 ± 0.04 ^b	2.14 ± 0.04 ^b
	43–49	1.93 ± 0.02 ^a	2.08 ± 0.03 ^b	2.08 ± 0.01 ^b	2.08 ± 0.02 ^b	2.06 ± 0.03 ^b	2.04 ± 0.03 ^b
	1–49	1.85 ± 0.02 ^a	2.11 ± 0.03 ^b	2.11 ± 0.04 ^b	2.12 ± 0.03 ^b	2.10 ± .04 ^b	2.11 ± 0.05 ^b

Means within a row with different superscript letters (a–d) denote significant differences ($P < 0.05$)

NC normal control, HSC heat stress control

addressed oxidative/nitrosative stress biomarkers adequately. Therefore, this study measured growth performance and several indicators of oxidative/nitrosative stress to investigate the protective effects of CEO against heat stress and provide an insight into the antioxidant properties of CEO compared to a standard commercial antioxidant.

As expected, heat stress negatively influenced performance of the broilers in this study. The overall weight of the birds under heat stress was at least 400 g lower than those of NC.

This finding is consistent with the study conducted by Tekke and Gül (2016) who observed decreased performance of broilers raised at 36 °C compared to those raised at 22 °C. It was determined that CEO acted better than vit E in increasing BW, BWG, and FI. In fact, there was no significant difference between ST and HSC; however, HT with the middle dose (350 ppm) could significantly increase BW, BWG, and FI compared to HSC. Consistent with these results, Habibi et al. (2014) demonstrated that supplementation of broilers

Table 4 Effects of different levels of clove essential oil and vit E on superoxide dismutase (SOD), glutathione peroxidase (GSH-Px), and catalase (CAT) activities, as well as malondialdehyde (MDA), protein carbonylation (PCO), nitric oxide (NO), and total antioxidant capacity (TAC) in broilers

	NC	HSC	100 ppm vit E	250 ppm CEO	350 ppm CEO	450 ppm CEO
RBC-SOD	473.42 ± 8.98 ^a	568.91 ± 9.75 ^b	552.82 ± 11.96 ^b	549.47 ± 10.07 ^b	539.82 ± 14.94 ^b	536.67 ± 14.46 ^b
RBC-GSH-Px	185.64 ± 3.68 ^a	249.53 ± 5.11 ^b	241.19 ± 4.79 ^b	242.12 ± 3.59 ^b	234.73 ± 6.21 ^{bc}	227.55 ± 4.21 ^c
RBC-catalase	92.79 ± 5.46 ^a	153.36 ± 2.94 ^b	142.36 ± 3.54 ^c	143.61 ± 2.27 ^c	137.91 ± 3.38 ^c	119.82 ± 1.19 ^d
Plasma-MDA	1.34 ± 0.02 ^a	2.52 ± 0.03 ^b	2.12 ± 0.05 ^c	2.14 ± 0.07 ^c	1.73 ± 0.05 ^d	1.42 ± 0.07 ^a
Plasma-PCO	1.77 ± 0.08 ^a	2.48 ± 0.04 ^b	2.15 ± 0.04 ^c	2.17 ± 0.6 ^c	1.96 ± 0.05 ^d	1.85 ± 0.07 ^{ad}
Plasma-NO	0.89 ± 0.03 ^a	1.36 ± 0.02 ^b	1.31 ± 0.03 ^b	1.32 ± 0.04 ^b	1.15 ± 0.08 ^c	0.97 ± 0.07 ^{ac}
Plasma-TAC	1.83 ± 0.09 ^a	0.76 ± 0.03 ^b	0.98 ± 0.06 ^c	1.05 ± 0.05 ^c	1.29 ± 0.08 ^d	1.64 ± 0.09 ^a

Means within a row with different superscript letters (a–d) denote significant differences ($P < 0.05$)

Enzymatic activity (SOD, CAT, GSH-Px) = U/gHb; MDA, PCO, and NO = nmol/L, TAC = milli mol/L

NC normal control, HSC heat stress control

under heat stress with different concentrations of ginger (*Zingiber officinale*) root powder and its EO could increase the overall BW and improve growth rate equally or better than 100 mg/kg of vit E, although there were no significant differences between the treatments and control group (Habibi et al. 2014). As mentioned, clove is a versatile plant and its antioxidant, anti-inflammatory, and antimicrobial properties have previously been proved (Batiha, Alkazmi, Wasef, Beshbishy, Nadwa and Rashwan 2020). Therefore, it is expected that CEO can increase performance of broilers through reducing the harmful bacterial populations of intestines, protecting amino acids against oxidation, and improving the function of digestive enzymes.

Antioxidant enzymes, most importantly SOD, GSH-Px, and CAT, tightly work together to protect tissues and organs against oxidative damages induced by free radicals and ROS. The physiologic concentration of these enzymes is tightly regulated and even a trifle alteration in the concentrations may result in abnormal function of the whole system leading to the vulnerability of biomolecules to oxidative injuries (Nazarizadeh and Asri-Rezaie 2016). The results of this study revealed that the activities of all the three enzymes increased in the birds under heat stress compared to the NC group. In agreement, Altan et al. (2003) observed increased activities of SOD, GSH-Px, and CAT in broilers (both Ross and Cobb strains) challenged with heat compared to the control birds under normal condition. The elevated activities of the enzymes can be due to induction of severe oxidative stress. In a compensatory mechanism, the activities have been increased to cope with the oxidative stress and alleviate harmful effects of oxidant molecules. Both vit E and CEO due to possessing direct antioxidant properties could reduce the activities and contribute to the normalization of the antioxidant enzymes. However, considering that SOD has the largest catalytic efficiency of any known enzyme (Nazarizadeh and Asri-Rezaie 2016), it plays a bigger role in the mitigation of oxidative stress compared to the other enzymes and its activity was not significantly influenced by the treatments. Perhaps, the level of oxidative stress was still high enough after supplementation with vit E and CEO to keep the activity of SOD elevated.

The status of oxidative stress can be ideally shown by measurement of lipid peroxidation and the assessment of MDA is a straightforward and reliable method to achieve this aim (Nazarizadeh and Asri-Rezaie 2016). Also, the degrees of protein damages induced by oxidant molecules can be assessed by measuring PCO. The results of this study clearly indicate that CEO can protect both lipids and proteins against oxidative injuries. Consistently, Petrovic et al. (2012) have measured increased levels of sulfhydryl groups and decreased levels of MDA in plasma samples of broilers treated with clove (Petrovic, Marcincak, Popelka, Simkova, Martonova, Buleca, Marcincakova, Tuckova, Molnar and Kovac 2012).

A significant decrease in NO concentrations following treatment with CEO was noticed in this study. NO is a key participant in many physiological pathways; however, its reactivity gives it the potential to cause considerable damage to the cells and tissues in its vicinity. Once produced, subsequent conversion of nitric oxide to nitrous anhydride and/or peroxynitrite can lead to the nitrosative damages to biomolecules (Burney et al. 1999). The data of decreased MDA and PCO contents suggests that CEO is able to protect lipids and proteins against nitrosative stress. Measurement of TAC gives a comprehensive insight into the balance between oxidants and antioxidants. In other words, the capacity of known and unknown antioxidants and their synergies are revealed by assessing TAC (Nazarizadeh and Asri-Rezaie 2016). Induction of heat stress could reduce TAC levels more than twofold compared to NC, suggesting that the antioxidant system of the birds was overwhelmed by oxidant molecules and free radicals. However, CEO in a dose-dependent fashion could efficiently restore TAC levels as there is no difference between the birds treated with 450 ppm CEO and NC animals. The improved performance and antioxidant status of the heat-challenged broilers can be related to the active ingredients of CEO. As depicted in Table 2, almost 73.19% of CEO is composed of eugenol, followed by eugenol acetate (9.57%). It has been well documented that eugenol is a potent antioxidant. In this regard, Nagababu et al. (2010) assessed the antioxidant activity of eugenol using both in vitro and in vivo studies and noticed that the inhibitory activity of eugenol was about fivefold higher than that of observed for alpha-tocopherol. It also could protect rat liver against carbon tetrachloride-induced oxidative damages (Nagababu et al. 2010). Additionally, Oso et al. (2019) observed that dietary supplementation with a phytogetic blend containing eugenol, methyl eugenol, and iso-eugenol could significantly improve growth performance, apparent ileal digestibility of nutrients, intestinal morphology, and cecal microflora of broiler chickens (Oso et al. 2019).

According to the obtained results, CEO acted better than the commercial antioxidant in the alleviation of adverse effects of heat stress. In a similar manner, Selvam et al. (2018) showed that a type of polyherbal formulation containing *Emblia officinalis*, *Ocimum sanctum*, and *Withania somnifera* could more efficiently stand against heat stress compared to vit C (standard antioxidant) in broiler chicks (Selvam et al. 2018). Therefore, CEO can be considered a potential feed additive with considerable antioxidant properties for the management of heat stress.

Compliance with ethical standards

Conflict of interest The author declares that there are no conflicts of interest.

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